JINMING ZHAO

Sunnyvale, CA · LinkedIn

PROFESSIONAL SUMMARY

Applied Scientist at Amazon AGI, working on conversational speech foundation models. Research spans multilingual speech recognition, speech translation, and reward modeling, with broader interests in speech generation. Experienced in building production reward models, preference-based post-training, and scalable multilingual data generation systems.
Technical Skills: Python, PyTorch, Hugging Face, SpeechBrain, speech foundation models, speech recognition, speech translation, reward modeling, SFT, DPO, evaluation modeling, multilingual modeling, SQL, AWS.

EDUCATION & QUALIFICATIONS

Doctor of Philosophy - **Monash University (Clayton)** Jan 2021– Jun 2025
Computer Science

Master of Data Science - **Monash University (Caulfield)** Feb 2018 – Jul 2020
Advanced Data Analytics

Bachelor of Commerce - **Monash University (Clayton)** Jul 2012 – Jun 2015
Double major in Accounting and International Commerce

RESEARCH & INDUSTRY EXPERIENCE

Applied Scientist – **Amazon (AGI)** Dec 2024 – Current
- Led development of a multi-dimensional speech evaluation model for scalable conversational quality assessment; designed the training formulation and deployed the system into production workflows, enabling automated evaluation.
- Redesigned a manually executed DPO data-generation workflow to address scalability and reliability limitations, refactoring existing recipes into a staged pipeline with validation checkpoints that enabled reproducible large-scale multilingual data generation.
- Diagnosed DPO training underperformance and identified a previously overlooked high-impact hyperparameter through targeted ablations; the revised configuration improved nearly all evaluation metrics without additional data or architecture changes.
- Developed and deployed an 11-language Whisper-based nativeness regression model for speech evaluation; identified a projection-layer-only fine-tuning regime that outperformed the prior production model, improving AUC from 0.79 to 0.88.

Research Scientist Intern – **Meta (FAIR)** Aug 2023 – May 2024
- Led first-author work extending Meta's Massively Multilingual Speech project to previously unseen low-resource languages; developed an unsupervised zero-shot ASR adaptation method for scalable multilingual recognition with limited or no labeled data, reducing error rate by 50% relative to the prior state-of-the-art baseline. (Publication)

Research Assistant - **Monash University** July 2020 – Aug 2023
- Developed transformer-based adaptor methods to bridge speech and text representations, improving parameter efficiency and low-resource performance in speech translation. This work informed research directions later reflected in Meta's Seamless Translator.
- Designed multilingual data construction and training strategies for low-resource text-to-text machine translation as part of a DARPA-funded research project, achieving consistent improvements across benchmark evaluations.

Visiting Researcher - **NARA Institute of Science and Technology** Oct 2022 – Dec 2022
- Built the first large-scale aligned English–Japanese simultaneous interpretation corpus and training pipeline, improving translation quality and latency in simultaneous translation systems. (Publication)

Teaching Associate - **Monash University** Feb 2023 – Jun 2023

- (co-lectured & tutored) FIT5217 – Natural Language Processing

PUBLICATIONS

- Jinming Zhao, Vineel Pratap, Michael Auli. 2024. Scaling A Simple Approach to Zero-Shot Speech Recognition. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025.
- Jinming Zhao*, Yuka Ko*, Ryo Fukuda, Katsuhito Sudoh, Satoshi Nakamura. 2024. NAIST-SIC-Aligned: an Aligned English-Japanese Simultaneous Interpretation Corpus. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation*.
- Jinming Zhao, Gholamreza Haffari, Ehsan Shareghi. 2023. Generating Synthetic Speech from SpokenVocab for Speech Translation. In *Findings of the 2023 Conference of the European Chapter of the Association for Computational Linguistics, 2023.*
- Yang Hao, Jinming Zhao, Ehsan Shareghi, Gholamreza Haffari. 2023. Investigating Pre-trained Audio Encoders in the Low-Resource Condition. In *Proceedings of the 2023 Interspeech.*
- Jinming Zhao, Yang Hao, Ehsan Shareghi, Gholamreza Haffari. 2022. M-Adapter: Modality Adaptation for End-to-End Speech-to-Text Translation. In *Proceedings of the 2022 Interspeech.*
- Jinming Zhao, Yang Hao, Gholamreza Haffari, Ehsan Shareghi. 2022. RedApt: An Adaptor for wav2vec 2 Encoding - Faster and Smaller Speech Translation without Quality Compromise. In *Findings of the 2022 Conference on Empirical Methods in Natural Language Processing, 2022.*
- Jinming Zhao*, Tongtong Wu*, Guitao Wang*, Zhaoran Liu, Guilin Qi, Yuan-Fang Li, Gholamreza Haffari. 2022. Towards Relation Extraction from Speech. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, 2022.*
- Jinming Zhao*, Yang Hao*, Gholamreza Haffari, Ehsan Shareghi. 2022. Self-supervised Rewiring of Pre-trained Speech Encoders: Towards Faster Fine-tuning with Less Labels in Speech Processing. In *Findings of the 2022 Conference on Empirical Methods in Natural Language Processing, 2022.*
- Jinming Zhao, Philip Arthur, Gholamreza Haffari, Trevor Cohn, Ehsan Shareghi. 2021. It is Not as Good as You Think! Evaluating Simultaneous Machine Translation on Interpretation Data. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing, 2021.*
- Jinming Zhao, Ming Liu, Longxiang Gao, Yuan Jin, Lan Du, He Zhao, He Zhang, and Gholamreza Haffari. 2020. SummPip: Unsupervised Multi-Document Summarization with Sentence Graph Compression. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '20).*

OTHER RELEVANT INFORMATION

LANGUAGES: English (professional fluency), Mandarin (native)
INTERESTS: Reading, Travelling, Hiking, Running and Rock Climbing
REFERENCES: References available upon request